Mohamed El Khoudimi

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Software and quantitative systems engineering, with work across execution research, backend reliability, data validation and AI evaluation.

SELECTED EXPERIENCE

BMO Financial Group

Database Operations Analyst Co-op | AI & Software Engineering | September-December 2026 | Toronto

Current co-op in Database Reliability, focused on infrastructure automation, operational data and software engineering. Detailed accomplishments will be added as completed work is verified for public presentation.

AstraSage Consulting

Founder & Strategic Analyst | January 2023-present

- Scoped and delivered client workflows for research, document generation and intelligent processing, integrating APIs, prompt orchestration and iterative output evaluation.

- Owned implementation, documentation and client delivery alongside financial modeling, dashboards and analytical reporting.

NTT Corporation

Fraud Analyst & Financial Reporting Analyst | May 2022-September 2024

- Investigated 200+ financial and fraud cases and documented findings for escalation under service-level and compliance requirements.

- Automated recurring reporting with Python and standardized validation. Reported outcomes include approximately 60% less manual reporting effort and 35% improved cross-team handover efficiency.

SELECTED SYSTEMS & RESEARCH

Quantitative Strategy Engine

C++ event-driven execution-research engine with full-depth order books, FIFO queues, queue-aware limit fills and market fills that walk available depth. Arena allocation, an SPSC ring buffer and cross-platform determinism; 302 C++ and 258 Python tests reported in the project record.

Research validation includes CPCV and Deflated Sharpe, with negative strategy results retained. Paper-trading integration; no claim of profitable live deployment.

Agentic Markets

Built a research environment separating agent-reported performance from independently calculated trading P&L, with a four-rung condition ladder and bootstrap analysis. Validated the analysis pipeline on a deterministic mock provider; a separate committed episode records agreement between reported and independently calculated P&L.

Event-Driven Fraud Engine / AACP

Built a fraud arbitration control plane with event ingestion, dynamic threshold control, agent orchestration and human escalation. The project record reports approximately 395 requests/second sustained throughput in k6 load testing.

Market microstructure research

Avellaneda-Stoikov order-queue audit - replicated three published tables, built a queue-aware simulator and validated 2,110,860 exchange messages. Held-out economic evaluation remains pending.

Kyle's lambda - pre-registered research over 351 million raw trades. Forecastability held on four holdout pairs; the execution-benefit criterion did not pass. Preserved corrected and as-run scorecards.

EDUCATION & TOOLS

Computer Programming Diploma, Georgian College | January 2025-April 2027 (expected) | GPA 3.7/4.0

Mechanical Engineering studies, Université de Moncton | August 2020-May 2024 | GPA 3.4/4.0 | completed in French

C++ | Python | SQL | FastAPI | Pydantic | pandas / NumPy | Git | Docker | CMake | GoogleTest | pytest

English | French | Arabic